Water Quality Monitoring using IoT and TinyML

Presented by:

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INTRODUCTION

1. Water quality is a key indicator of environmental health and human well-being.
2. Traditional testing methods are slow, expensive, and rely on manual sampling and laboratory analysis.
3. There is a need for continuous, real-time, and automated monitoring systems.
4. IoT (Internet of Things) enables remote sensing by connecting various water quality sensors.
5. TinyML (Tiny Machine Learning) allows on-device analysis, reducing dependency on cloud computation.
6. Combining IoT + TinyML creates a smart water quality monitoring system capable of local decision-making and instant anomaly detection.
7. This system ensures efficient water management, pollution control, and early warning for contamination events.

OBJECTIVE

- To monitor essential water parameters (pH, temperature, turbidity, conductivity, and dissolved oxygen).
- To apply TinyML algorithms on microcontrollers for real-time data analysis.
- To reduce manual monitoring and water wastage through automation.
- To create a low-cost and energy-efficient water quality monitoring solution.
- To support sustainable smart city development and environmental protection.

PROBLEM STATEMENT

Traditional Challenges:

- Manual testing is time-consuming and prone to human error.
- Cloud-based analysis introduces delays and depends on internet connectivity.
- No immediate feedback or alerts for contamination.
- High operational costs in large-scale monitoring setups.

Proposed Solution:

A TinyML-enabled IoT water monitoring system that analyzes sensor data locally, detects pollution instantly, and provides alerts in real time — even in remote areas without stable connectivity.

Components Used

Hardware Components

- ESP32: Main controller; reads sensor data and sends it to cloud platforms.
- TDS Sensor: Measures dissolved solids to check water purity.
- LCD Display: Shows real-time TDS readings and water quality status.
- Breadboard & Jumper Wires: Used for easy circuit connections and testing.
- Power Supply (USB/Battery): Provides power for ESP32 and sensors.

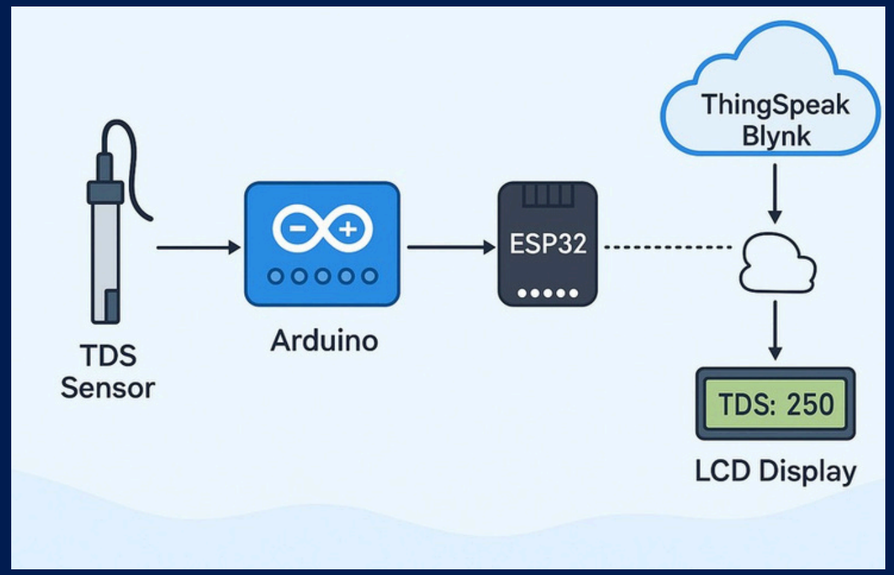
Software / Cloud Components

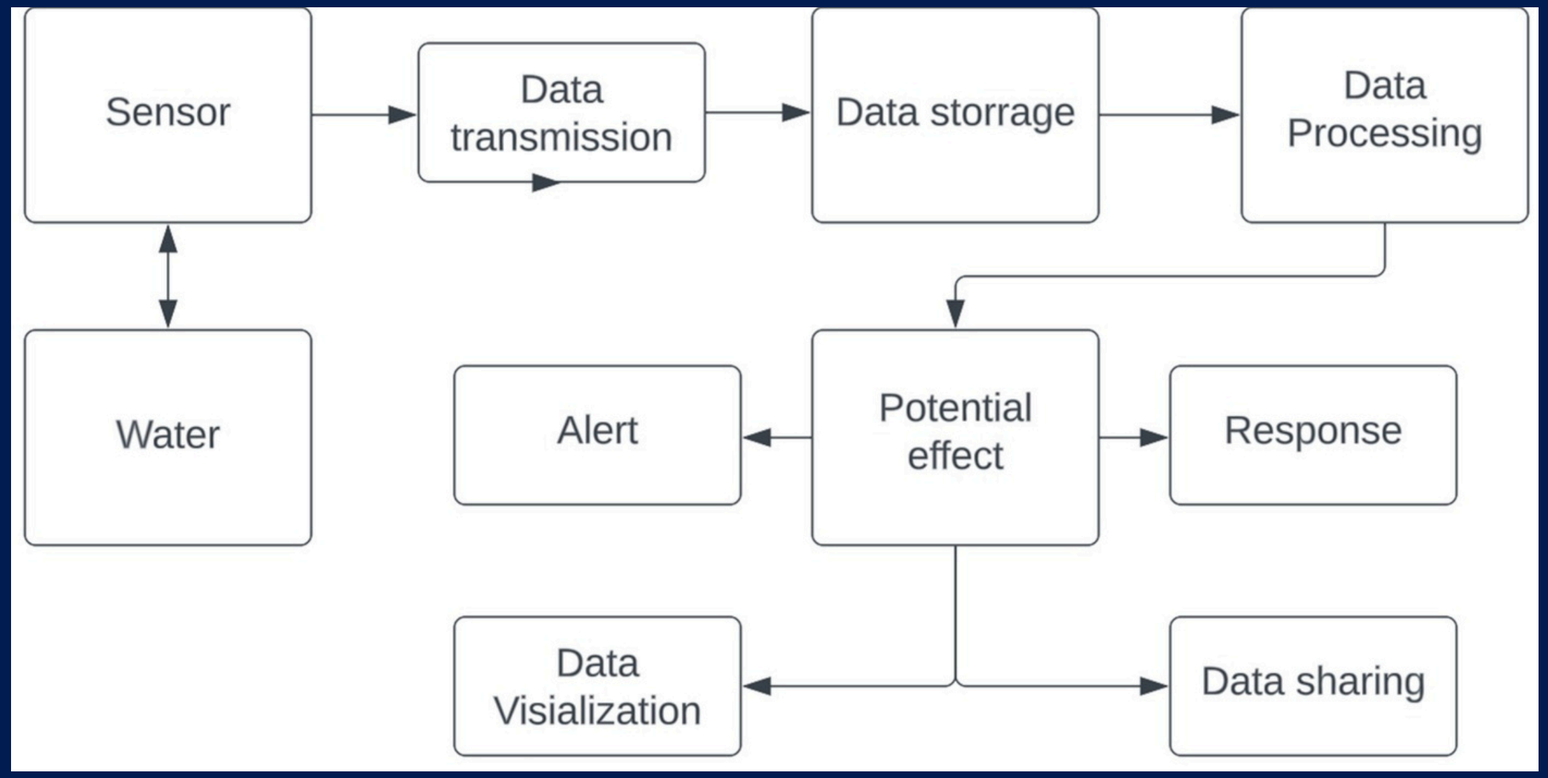
- ThingSpeak: Cloud platform for storing and graphing water quality data.
- Blynk App: Mobile dashboard for live monitoring and alerts.
- Arduino IDE: Used for programming and uploading code to ESP32.

SYSTEM OVERVIEW

- The IoT-based Water Quality Monitoring System is designed to measure and analyze TDS (Total Dissolved Solids) in water using Arduino and ESP32 microcontrollers.
- The TDS sensor detects the level of dissolved salts and impurities from the water sample and sends analog signals to the Arduino.
- The Arduino converts the sensor readings into a digital TDS value and transfers it to the ESP32 for further processing.
- The ESP32 performs TinyML or threshold-based analysis to classify the water as Pure, Moderate, or Contaminated.
- The processed data is shown on an LCD display for local viewing and simultaneously uploaded via Wi-Fi to ThingSpeak and Blynk for cloud visualization.
- Users can view live TDS readings, graphs, and alerts through the Blynk mobile app or ThingSpeak dashboard, enabling real-time water quality monitoring from anywhere.

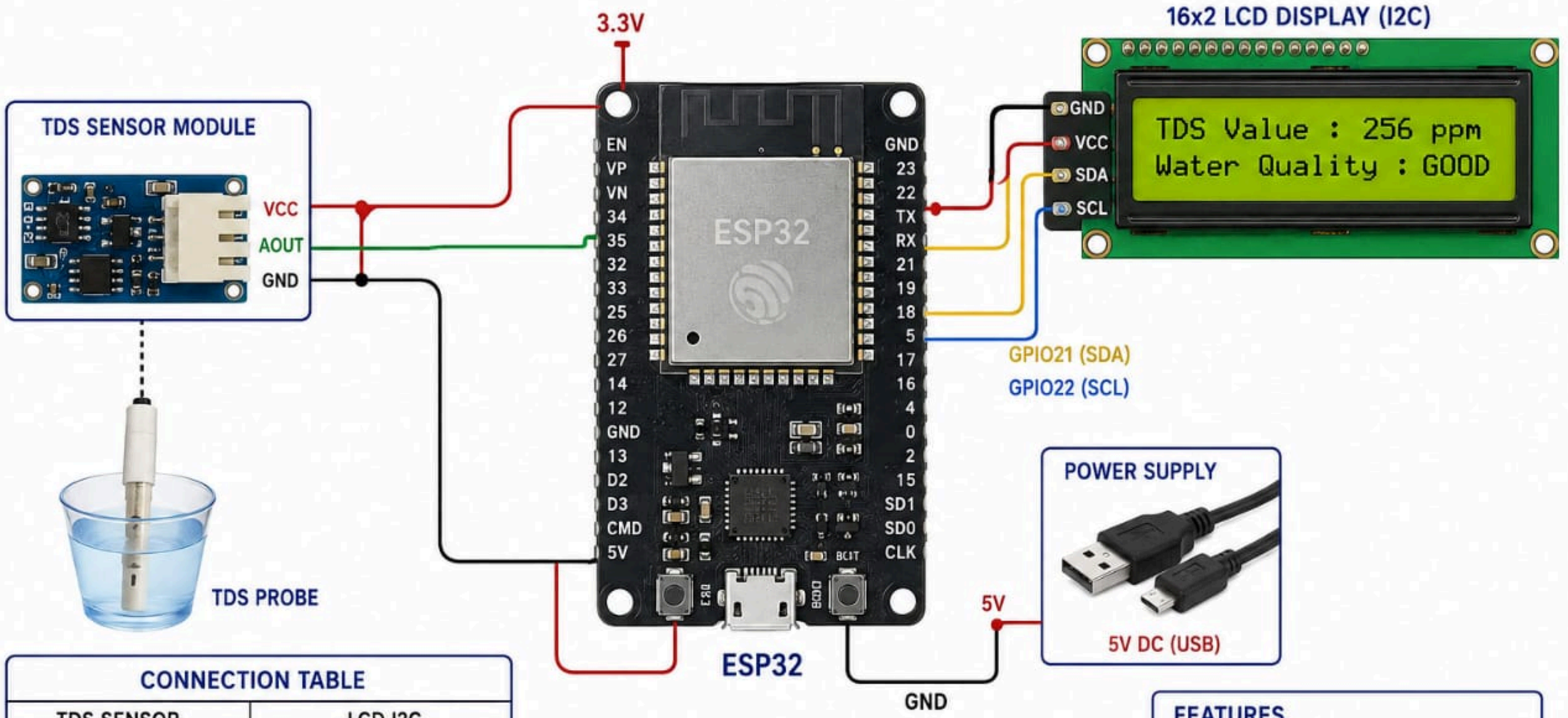
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SMART WATER QUALITY MONITORING SYSTEM

USING TDS SENSOR (WITHOUT ARDUINO)



CONNECTION TABLE

TDS SENSOR		LCD I2C	
VCC	3.3V	VCC	5V
GND	GND	GND	GND
AOUT	GPIO34	SDA	GPIO21
		SCL	GPIO22

FEATURES

- Measures TDS (ppm) in water
- Real-time monitoring on LCD Display
- Wi-Fi enabled (ESP32)
- Data can be sent to IoT platforms (ThingSpeak / Blynk / etc.)

WORKING PRINCIPLE

1. **Water Sampling**: The TDS sensor is placed in the water source to measure the concentration of dissolved solids (impurities, salts, minerals).
2. **Signal Acquisition (Arduino)**: The Arduino boardreads theanalogvoltagesignalfrom the TDS sensor. It converts this signal into a digital TDS value (in ppm).
3. **Data Processing (ESP32):** The ESP32 microcontroller receives the processed TDS data from Arduino.It performs local analysis or TinyML-based classification to determine whether the water is Pure, Moderate, or Contaminated.
4. **Display and Transmission:**
   - The result (TDS value and quality status) is shown on the LCD display.
   - The ESP32 sends the same data via Wi-Fi to ThingSpeak and Blynk cloud platforms.
5. **Cloud Visualization:**
   - ThingSpeak: Displays real-time and historical TDS data as interactive graphs.
   - Blynk App: Provides live readings and notifications on a mobile dashboard.
6. **Alert and Decision**: If TDS values cross the preset threshold, the system automatically triggers alerts through the Blynk app, enabling timely action.

Advantages

- Real-time water quality monitoring
- Low latency due to on-device TinyML processing
- Reduced dependence on cloud connectivity
- Energy-efficient and cost-effective
- Early detection of contamination and pollution
- Helps in sustainable water management and conservation

Challenges

- Sensor calibration and accuracy in diverse water environments
- Limited memory and power on microcontrollers
- Environmental noise affecting sensor readings
- Maintaining reliable connectivity in remote monitoring stations
- Data labeling and model training for diverse conditions

Outcomes & Results

- Achievedaccurateprediction of water quality (toxicity, pH imbalance, and contamination levels).
- TinyML-based inference achieved ~95–98% accuracy with low power consumption.
- Real-time alert system helped prevent water wastage and detect contamination early.
- Demonstrated scalability for use in smart cities, lakes, and industrial discharge monitoring.

Future Scope

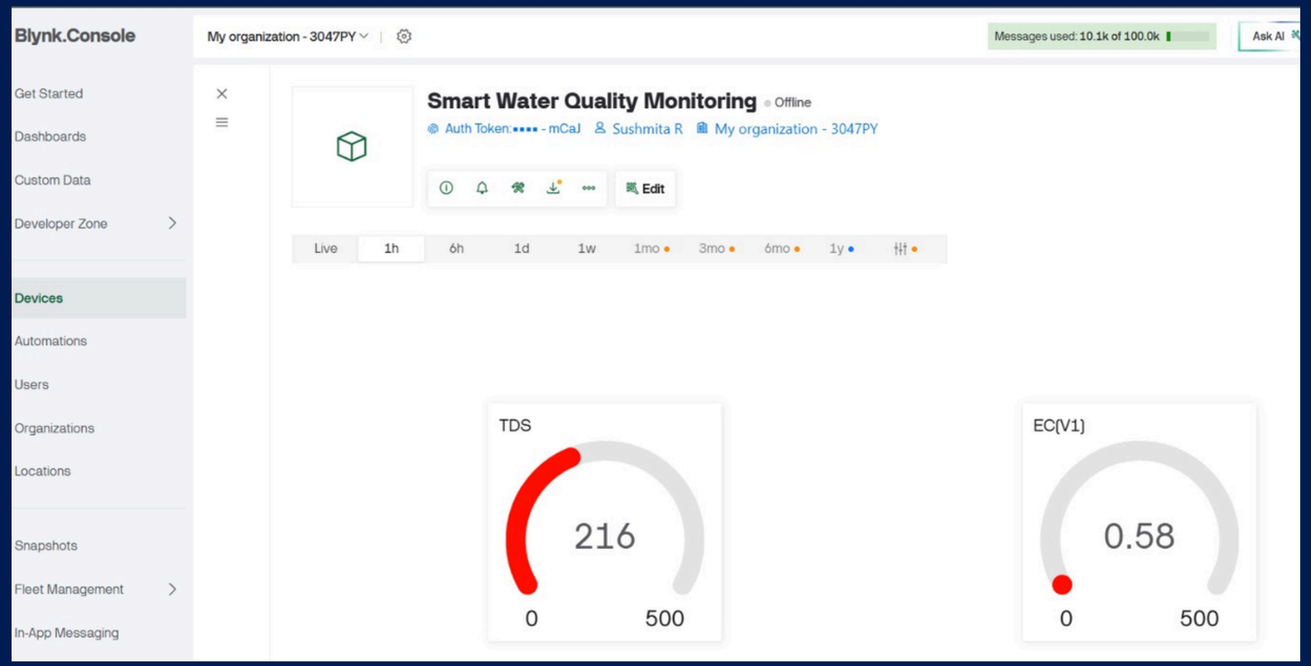
- Predictive analytics: Forecast long-term trends in water quality using advanced ML models.
- Integration with cloud AI: Combine edge and cloud learning for adaptive intelligence.
- Solar-powered monitoring stations for remote deployment.
- Automated purification systems triggered by contamination alerts.
- Expansion to wastewater and industrial monitoring.

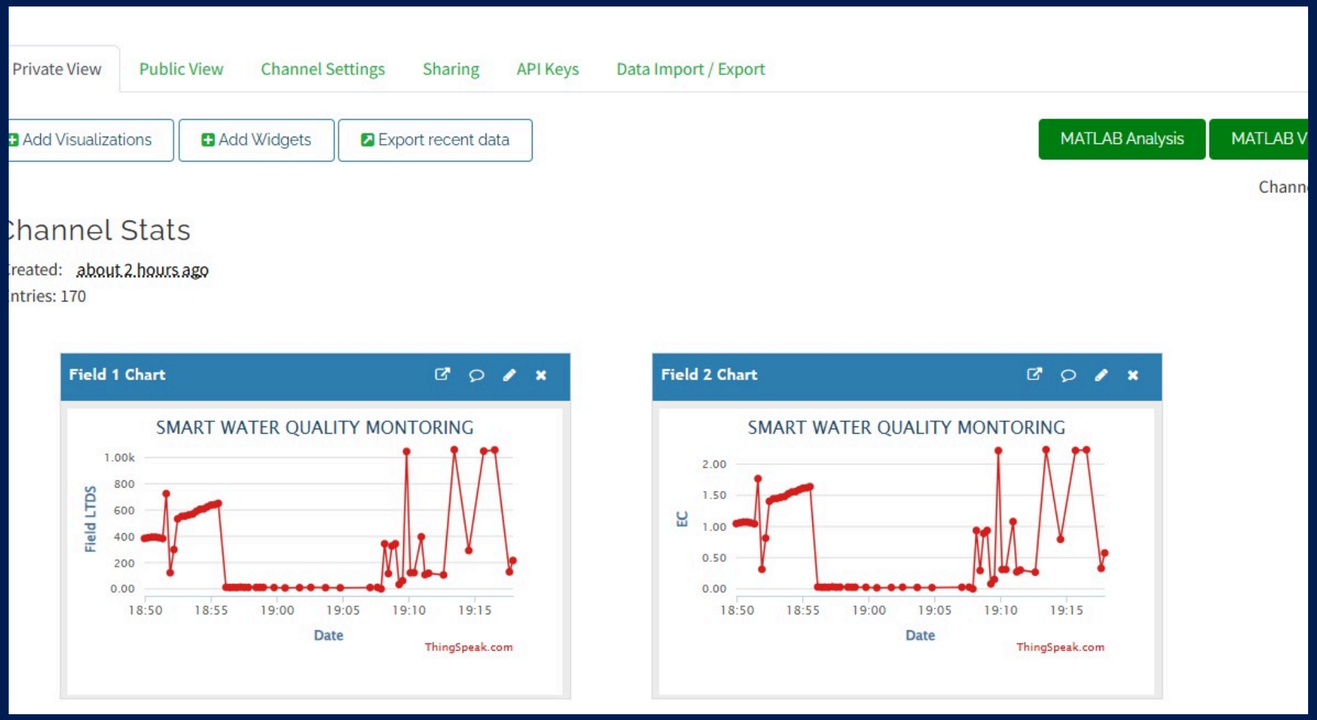
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Blynk.Console
My organization - 3047PY
Messages used: 10.1k of 100.0k
Ask AI
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Literature Survey

- IoT-based water monitoring systems are widely used for real-time tracking of parameters like pH, turbidity, and TDS using cloud platforms such as ThingSpeak and Blynk.
- Studies show ESP32 and Arduino microcontrollers are reliable for low-cost, continuous data collection and remote monitoring.
- Integration of TinyML enables local data analysis on microcontrollers, reducing cloud dependency and improving response time.
- Previous works highlight benefits like real-time alerts, low power use, and easy scalability for smart city and environmental applications.

Conclusion

- The proposed system efficiently monitors water quality using TDS sensor, Arduino, and ESP32, combined with ThingSpeak and Blynk for visualization and alerts.
- It provides real-time monitoring, remote access, and automatic detection of water purity levels.
- The design is cost-effective, energy-efficient, and supports smart and sustainable water management.
- Future enhancement: adding TinyML-based prediction and more sensors (pH, turbidity, DO) for broader water-quality analysis.

Thankyou!